
ECE228 Project Proposal: Air Quality Prediction

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1 Project Proposal: Track 2

1.1 Problem Background & Motivation

Air pollution, particularly fine particulate matter (PM_{2.5}), poses a serious threat to public health in dense urban areas. PM_{2.5} is especially harmful because fine particles can be inhaled deeply into the lungs and are associated with respiratory and cardiovascular risks. Accurate short-term air quality forecasting is therefore not only a prediction problem, but also an early-warning problem: if high PM_{2.5} levels can be predicted several hours in advance, cities can issue public health advisories and individuals can adjust outdoor activities.

The motivation of this project is to build a data-driven PM_{2.5} forecasting pipeline using real urban air-quality and meteorological data. PM_{2.5} prediction is challenging because air pollution is affected by nonlinear interactions among historical pollutant levels, weather conditions, and temporal patterns. For example, wind speed and wind direction influence pollutant dispersion, while temperature, pressure, humidity, and rainfall can affect pollutant accumulation and atmospheric stability.

A key challenge is that air quality is not purely an isolated single-station time series. In reality, pollution is a spatio-temporal physical process: pollutant concentration at one monitoring station may be related to nearby stations due to transport, diffusion, and shared meteorological conditions. Therefore, a model that only uses the historical values of one station may miss useful contextual information from other sites.

We plan to use the Beijing Multi-Site Air Quality Dataset from the UCI Machine Learning Repository. This dataset contains synchronized hourly air-quality and meteorological measurements from 12 monitoring sites in Beijing. It includes six major pollutants, including PM_{2.5}, PM₁₀, SO₂, NO₂, CO, and O₃, as well as meteorological variables such as temperature, pressure, dew point, rainfall, wind direction, and wind speed. This makes the dataset suitable for studying both temporal prediction and simple multi-site modeling. Our project aims to investigate whether sequence modeling and multi-site contextual features can improve short-term PM_{2.5} forecasting compared with standard feature-based regression models.

1.2 Related Work

Prior research on air quality prediction can be organized into three main categories.

1. Statistical and physical forecasting models. Early approaches include statistical time-series models and chemical transport models. Statistical models such as ARIMA are relatively interpretable and can capture short-term temporal patterns, while chemical transport models incorporate physical and chemical knowledge about pollutant formation and transport. However, these approaches may struggle to capture complex nonlinear relationships among pollutants, weather variables, and urban dynamics, especially when the input data are high-dimensional and collected from multiple monitoring stations.

2. Classical machine learning and temporal sequence models. To model nonlinear relationships, many studies use machine learning models such as linear regression, support vector regression, random forest, and gradient boosting. These methods can combine pollutant measurements, meteorological variables, and time features, making them strong baselines for PM_{2.5} prediction. However, standard machine learning models often treat each training example independently. Sequence models such as recurrent neural networks, GRU, and LSTM address this limitation by explicitly learning

from historical time windows. They are well suited for hourly air-quality prediction because current PM2.5 concentration is strongly related to recent historical pollution and weather conditions.

3. Spatio-temporal deep learning. Recent work has increasingly focused on modeling both temporal dynamics and spatial correlations across monitoring stations. For example, GeoMAN uses multi-level attention to predict geo-sensor readings by considering sensor readings, meteorological data, and spatial information. Other work combines graph-based models with sequence models to represent spatial relationships between monitoring sites. These methods motivate our project because the Beijing dataset contains multiple synchronized monitoring stations, making it possible to explore whether multi-site context improves prediction.

Our project is positioned between classical machine learning and advanced spatio-temporal learning. We do not aim to reproduce a full state-of-the-art graph neural network system. Instead, we will build a practical and reproducible PM2.5 prediction pipeline. We will compare non-sequential baselines with sequence-based models, and we will also test whether adding multi-site contextual features improves short-term forecasting performance.

1.3 High-level Methodology

The main idea of this project is to build an end-to-end short-term PM2.5 forecasting pipeline. Given recent historical air-quality and meteorological measurements, the model will predict future PM2.5 concentration, such as one hour ahead or several hours ahead.

First, we will preprocess the Beijing Multi-Site Air Quality Dataset by sorting records by station and time, handling missing values, normalizing numerical variables, and encoding categorical variables such as wind direction and station ID. We will also create time-based features such as hour of day, day of week, and month to capture daily and seasonal patterns.

Second, we will construct supervised learning examples using a sliding-window approach. For each station, the input may contain the previous 24 hours of pollutant and meteorological measurements, and the target will be the future PM2.5 value. To study the value of multi-site information, we will experiment with two input settings: a station-only setting that uses data from a single target station, and a multi-site setting that adds contextual features from other monitoring stations or trains one global model across all stations with station ID as an input feature.

Third, we will train and compare several models. As baselines, we will use linear regression and a tree-based model such as random forest or gradient boosting. These models will show how much performance can be achieved from feature-based learning. Then, we will train sequence models such as GRU or LSTM to capture temporal dependencies across recent hours. If time allows, we may also test a lightweight attention-based model to examine whether the model can focus on the most relevant time steps or features.

Finally, we will evaluate the models using standard regression metrics, including mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). In addition to average prediction accuracy, we will visualize model performance around high-pollution periods, since forecasting sudden PM2.5 spikes is important for early warning. The expected outcome is a reproducible air-quality prediction pipeline and an analysis of whether temporal modeling and multi-site context improve short-term PM2.5 forecasting.

References

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